

Open by design **Open science**



Open infrastructure for equitable cooperation

How Brazilian public infrastructure can advance a shared Global South Open Science agenda.

Washington Segundo

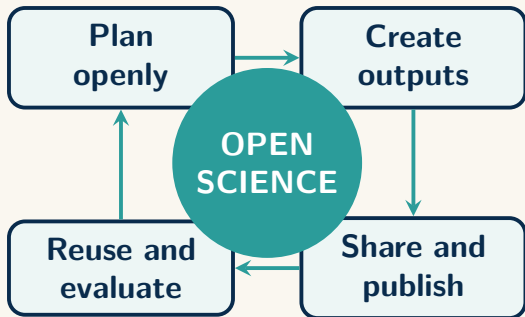
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Ministry of Science, Technology and Innovation



Open science redesigns the entire research cycle



A paradigm shift

- Openness begins at planning, not after publication.
- Publications, data, code, methods, and reviews become reusable outputs.
- Governance, preservation, and participation are part of scientific quality.

Core idea: knowledge should be accessible, auditable, interoperable, and useful to society.

A long evolution led from private exchange to open ecosystems



Digital networks separated the traditional functions of journals: record, certification, dissemination, and preservation.

Ibict has shaped scientific information

1954

Founded as Brazil's national
institute dedicated to
bibliography and scientific
documentation

Today

A research unit of the
Ministry of Science,
Technology and Innovation
- MCTI

A long-standing public mission

- Organize, preserve, and disseminate Brazil's scientific and technological information.
- Build national services, standards, and interoperable information infrastructures.
- Connect Brazilian research with regional and international knowledge networks.

A strong tradition of openness Ibict has played a sustained leadership role in **Open Access** and **Open Science**, advancing repositories, aggregators, research data, digital preservation, persistent identifiers, and public-interest governance.

Four axes turn information into public value

Information integrity and justice

Preserve authorship and meaning across the information cycle while ensuring equitable production, access, and use.

Information sovereignty

Strengthen public control over information flows, service infrastructures, algorithms, standards, software, and hardware.

Information for citizenship

Build critical, media, and data literacy for a multilingual, accessible, and socially inclusive information environment.

Open Science

Open research practices strengthen public trust, cooperation, and science diplomacy.

Information Science connects the four axes to support social, environmental, and economic development.

Source: Braga, *Ciência da Informação*, vol. 55, no. 1, 2026.

Open infrastructures connect research to the SDGs

SDG 4: Quality education

Open knowledge, education and training, and skills development.

SDG 9: Industry, innovation and infrastructure

Interoperable public systems for research and innovation.

SDG 10: Reduced inequalities

More equitable participation and visibility across communities.

SDG 16: Peace, justice and strong institutions

Evidence for policy monitoring and public accountability.

SDG 17: Partnerships for the goals

Shared standards, tools, and institutional cooperation.

Social technologies connect research with locally defined public needs.

Proposed pathways to public benefit, with outcomes to be evaluated.

Source: United Nations Sustainable Development Goals.

A Global South action plan turns principles into action

African Union · Brazil · China · South Africa

Shared principles: equality and mutual benefit | respect for diversity
universal access | open collaboration

1 Global governance

Advance UNESCO principles through joint governance and monitoring.

2 Open data cooperation

Connect platforms for ethical and sustainable sharing.

3 Shared infrastructure

Co-develop public services and trusted data spaces.

4 Open scholarly communication

Share papers and preprints through inclusive infrastructures.

5 Public engagement

Build culture, education, communication, and participation.

6 Skills in developing countries

Develop skills through exchange and open technology.

The ecosystem is broader than open access

Open outputs

Publications, research data, software, methods, and educational resources.

Open processes

Preprints, transparent peer review, responsible assessment, and reproducible workflows.

Open infrastructures

Repositories, identifiers, metadata standards, preservation, and research information systems.

Open participation

Citizen science, knowledge dialogue, inclusion, and public engagement.

Infrastructure connects every pillar and keeps openness sustainable.

Four principles balance technical value and social responsibility

FAIR

Make data findable, accessible, interoperable, and reusable for people and machines.

TRUST

Build transparent, responsible, user-focused, sustainable, and technically sound infrastructures.

CARE

Protect collective benefit, authority, responsibility, and ethics in data governance.

DEIA

Embed diversity, equity, inclusion, and accessibility in research systems.

Rule of thumb: as open as possible, as closed as necessary.

Maximize openness effectively, strategically, inclusively, and responsibly

Open publishing must close the scientific information divide

Route	How it opens access	Main strength	Main risk
Green	Deposit in a repository	Uses public infrastructure	Embargoes or restrictive publisher policies
Gold	Immediate access on the publisher platform	Open version of record	APCs, when charged, can exclude authors
Diamond	No fees for readers or authors	Aligns publishing with the public interest	Sustained funding and governance

Diamond publishing needs sustained funding, shared governance, and strong editorial standards.

Publish, review, curate: the power of preprints

1. Publish

Share a preprint early, with a persistent identifier and clear version history.

2. Review

Invite critical evaluation and make reviews and author responses discoverable.

3. Curate

Select, contextualize, and recommend work using transparent editorial criteria.

- Earlier access can accelerate feedback, collaboration, and reuse.
- Link preprints, reviews, revised versions, and journal publications.
- Display review status clearly. Early availability does not establish validity.

Source: COAR and ASAPbio, Publish, Review, Curate initiative.

Openness must also deliver informational justice

Three dimensions

- **Distribution:** equitable access to information and infrastructure.
- **Recognition:** respect for diverse people, knowledge, and epistemologies.
- **Participation:** meaningful inclusion in producing and using knowledge.

Ethical safeguards

- Consent and risk assessment
- Privacy and controlled access
- Collective and Indigenous data rights
- Protection from harmful or dual use

Implementation depends on human mediation

Inclusive design Build for diverse communities

- Accessible interfaces
- Multilingual discovery
- Disciplinary flexibility
- Context-sensitive workflows

Governance Define responsibility

- Data control and privacy
- Curation criteria
- Ethical safeguards
- Sustainable ownership

Skills development Enable critical use

- Education and training
- Limits and uncertainty
- Reproducible practices
- Recognition of data and software

Reward data sharing, software, review, and curation. Support teams turn principles into practice.

Brazil connects public services across the research cycle

Collect, publish, and connect

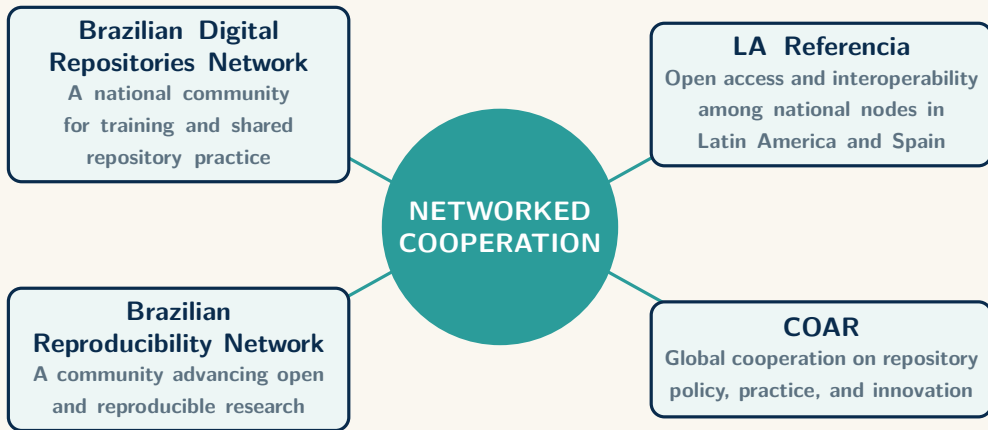
- SciELO, journals, and institutional repositories
- BDTD for graduate research
- Oasisbr for discovery across collections

Integrate, identify, and preserve

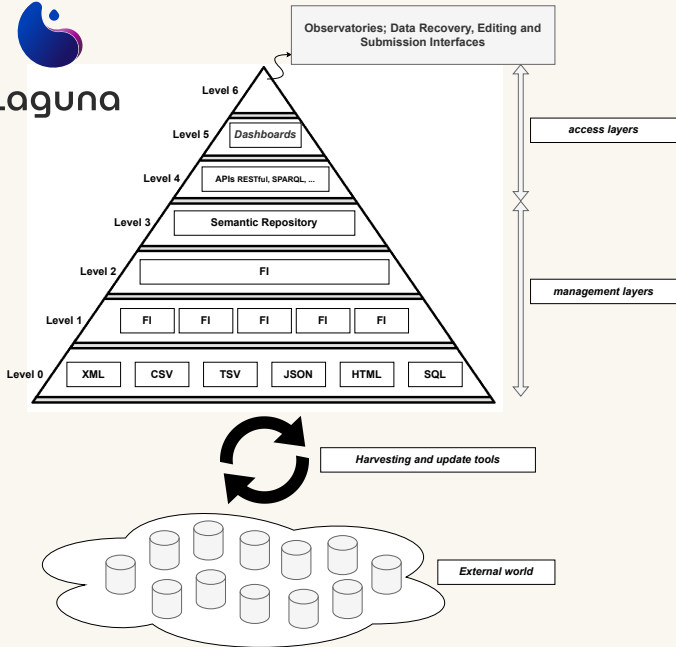
- BrCris for connections across the research system
- Laguna for heterogeneous data integration
- dARK for identifiers and Cariniana for preservation

Shared standards and stable funding support interoperability and institutional autonomy.

Networks connect national expertise with global cooperation



Networks operationalize Objectives 1 and 3 through shared standards, learning, and collective action.

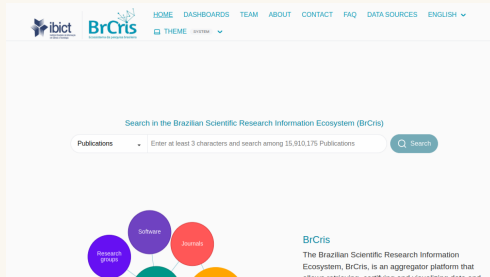


BrCris connects the Brazilian research ecosystem

A national research information system

- Connect researchers, institutions, projects, and outputs.
- Explore relationships to support research management.
- Investigate further interoperability with OpenAlex and OpenCitations.

15.9M publications: platform count displayed in July 2026.



Further integrations are a co-operation opportunity to assess.



Oasisbr connects more than 6.5 million open digital objects



A national and international gateway

6.5M+

articles, theses, dissertations, books, book chapters, and research datasets

~1.2M

documents aggregated through BDTD

~2,000

information sources: repositories and scientific journals across Brazil



BDTD gives visibility to graduate research

A national service since 2002

- Conceived, developed, and maintained by Ibict.
- Integrates theses and dissertations from Brazilian teaching and research institutions.
- Also includes records of works defended by Brazilians at institutions abroad.
- Offers free national discovery; full texts remain hosted by participating institutions.

Public value

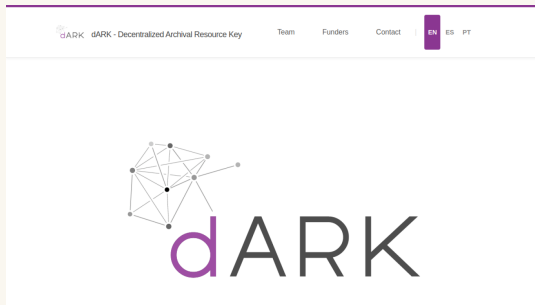
- Visibility for postgraduate research
- Shared metadata standards
- Institutional interoperability
- Access to specialized knowledge
- Accountability for public investment



dARK connects persistent identification and shared governance

Institutional responsibility

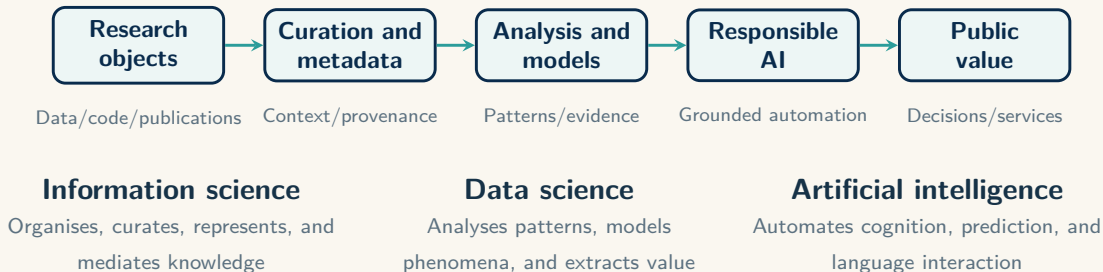
- Decentralized identifiers based on ARK
- Traceable records and shared operational roles
- Governance, funding, and maintenance designed from the start



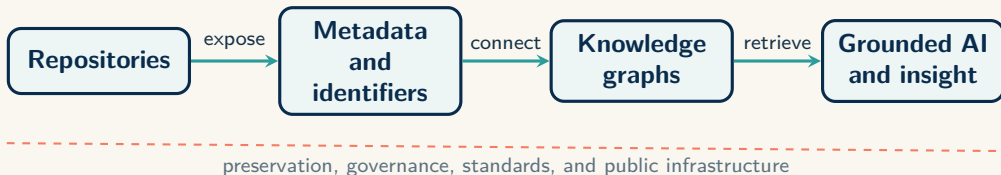
LA Referencia collaboration, with support from Invest in Open Infrastructure.

Identification supports traceability. Preservation keeps the underlying resources available.

Information science is the foundation of reliable AI



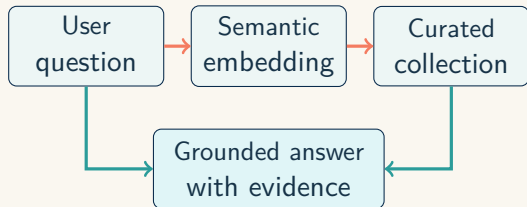
Trustworthy AI begins with trustworthy research infrastructure



Preservation and interoperability turn stored objects into reusable evidence and enable the trusted data spaces and AI applications called for in Objective 3.

Semantic systems connect meaning, not only keywords

A grounded retrieval flow



What makes it work

- Controlled vocabularies and ontologies reduce ambiguity.
- Multilingual embeddings retrieve concepts across languages.
- RAG links answers to retrieved documents.
- Check sources, omissions, and uncertain answers.

Applied AI creates value when the information base is ready

Application	AI capability	Public-interest value
ServicesPI	Semantic search + NLP	Natural-language discovery across complex patent collections
Patente.IA	Text + image similarity	Broader detection of related inventions and prior art
RDAPP	NLP + semantic retrieval	Structured access and analysis across public-policy documents
SAVITS	BI + ML + LLM	Monitoring and analysis of social technology policies

Public control of models and infrastructure requires skills development, source checking, and human oversight.

Ibict can contribute across all six objectives

Governance, data, infrastructure

- **1 Governance:** policy expertise, networks, standards, and monitoring.
- **2 Open data:** Oasisbr, BDTD, metadata aggregation, and interoperability.
- **3 Infrastructure:** BrCris, Laguna, Cariniana, dARK, and public digital services.

Communication, participation, skills

- **4 Communication:** repositories, journals, open publishing, and persistent access.
- **5 Engagement:** information literacy, citizen science, and multilingual access.
- **6 Skills:** training, South–South exchange, and open technology transfer.

Brazil can move from endorsing shared principles to delivering interoperable services, professional skills, and evidence for cooperation.

Mapped to the Action Plan for International Cooperation in Open Science, 2025.

National delivery enables international cooperation

National backbone

- A focal point linking domestic and international partners
- Defined roles for government, funders, institutions, and networks
- Stable funding for public infrastructures and professional teams
- Shared standards with regional and disciplinary adaptation

Multilateral mechanism

- An exchange mechanism with a defined added value
- Coordinated resource allocation and practical pilot projects
- Dynamic monitoring and transparent reporting
- Rotating meetings to share evidence and improve practice

Strong national institutions are the foundation for equal participation, joint contribution, and shared benefits.

A joint pilot for policy monitoring and fair research assessment

Public policy monitoring

- Start with policies related to social technologies.
- Track implementation, reach, and reported outcomes.
- Interpret evidence with managers and participating communities.

Research assessment tools

- Connect repositories, journals, and research information systems.
- Document diverse contributions and their social use.
- Support contextual assessment by researchers and reviewers.

Proposal for co-design: agree partners, responsibilities, resources, safeguards, and a timetable.

Connected evidence can support policy and assessment tools

Evidence to connect	Proposed tool	Use to test
Publications, data, and software	Links between repositories, journals, and research records	Trace contributions and reuse across outputs
Policy documents and project reports	Policy monitoring dashboard with source links	Examine implementation, coverage, and evidence gaps
Reviews and community accounts	Contextual evidence portfolios for qualitative review	Examine social use, local relevance, and differing perspectives

Start with an agreed sample. Document provenance, permissions, uncertainty, and correction procedures.

CoARA provides a reference for fairer research assessment

Assessment principles

- Recognize diverse outputs, activities, roles, and languages.
- Prioritize qualitative judgement with peer review.
- Use quantitative indicators responsibly and in context.

Implications for the pilot

- Avoid inappropriate use of journal metrics and the h-index.
- Develop criteria with researchers and assessors.
- Test tools for bias, transparency, and practical usefulness.

Coalition for Advancing Research Assessment. Reference for the proposal, without implying institutional membership.

Source: CoARA commitments, especially 1–3, 6, 7, and 10.

What would success look like?

Intended result	Evidence to collect	Review question
Useful policy monitoring	Documented use by managers and communities	Did the evidence inform action?
More inclusive assessment	Contributions recognized across languages, roles, and outputs	Whose work became visible?
Traceable social contribution	Cases of use, participant accounts, and contextual review	What changed, for whom?
Sustainable shared tools	Reuse, maintenance roles, and demonstrated skills	Can partners sustain the work?

Agree a baseline and review criteria before the pilot. Assess contribution to outcomes without assuming causality.

Brazil can help turn the action plan into shared infrastructure



Connect services + launch pilots + develop skills + share evidence

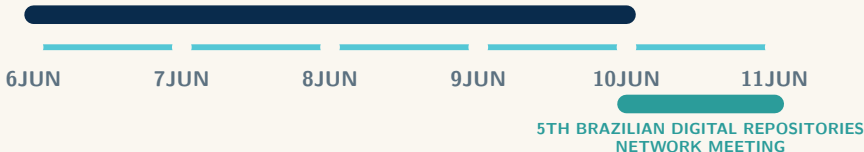
Ibict can contribute services and partnerships to policy monitoring and fairer research assessment, with social benefit as a shared purpose.

Salvador/Brazil connects repository communities in 2027



OpenRepositories

OPEN REPOSITORIES 2027



6–10 JUNE 2027

Open Repositories 2027

Salvador, Bahia: global exchange comes to South America's repository community.

10–11 JUNE 2027

Brazilian Repository Network

The fifth national meeting advances cooperation and shared repository practices.

A 2027 milestone for shared pilots, progress review, and network alignment.



Thank you!

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Aguyjevete

Guarani

Ẹ sẹ́

Yoruba

Sulpayki

Quechua

Na gode

Hausa

Asante

Swahili / Bantu

धन्यवाद

Hindi

Ngiyabonga

Zulu / Bantu

ধন্যবাদ

Bengali

நன்றி

Tamil

አመሰግናለሁ

Amharic

Terima
kasih

Indonesian

谢谢

Chinese

Obrigado

Portuguese

